

2017 HKDSE Physics

Paper 1A — Multiple-choice Questions

Original questions and official answers with detailed English reasoning. Calculations show relationships, substitution and units; each structured subpart is presented separately.

Question 1

30 g of milk at 10 °C is added to 120 g of coffee at 80 °C. Assuming there is no heat loss to the surroundings, what is the final temperature of the mixture?

Given: specific heat capacity of milk = $3800 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$

specific heat capacity of coffee = $4200 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$

A. 64.8 °C

B. 65.2 °C

C. 66.0 °C

D. 67.1 °C

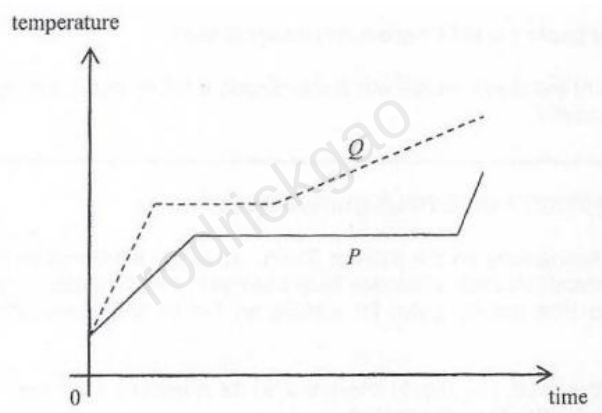
Official answer: D

With no heat loss, heat lost by coffee equals heat gained by milk: $0.120(4200)(80-T) = 0.030(3800)(T-10)$.
Hence $T = (504 \times 80 + 114 \times 10) / (504 + 114) = 67.1 \text{ }^{\circ}\text{C}$. Weight temperatures by heat capacity mc, not by mass alone. Answer: D.

Question 2

Same mass of solids P and Q are heated at the same rate. The temperature-time graphs of the two substances are shown below.

Which of the following comparisons about their melting points and specific latent heats of fusion is correct?



- A. Higher melting point: P; larger specific latent heat of fusion: P
- B. Higher melting point: P; larger specific latent heat of fusion: Q
- C. Higher melting point: Q; larger specific latent heat of fusion: P
- D. Higher melting point: Q; larger specific latent heat of fusion: Q

Official answer: C

The horizontal section gives the melting point, so Q has the higher melting point. During melting, $P_t = mL$.

Equal masses and equal heating powers mean latent heat is proportional to the plateau duration. P has the longer plateau and therefore the larger specific latent heat of fusion. Answer: C.

Question 3

Which of the following statements about the internal energy of a substance are correct?

- (1) When a solid melts, the latent heat of fusion absorbed becomes potential energy of the molecules in the substance.
- (2) When a vapour condenses, its internal energy decreases.
- (3) When a liquid evaporates, the internal energy of the remaining liquid increases.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: A

During melting at constant temperature, absorbed energy increases molecular potential energy rather than average kinetic energy. Condensation releases energy, so internal energy decreases. In evaporation, higher-energy molecules escape; without compensating heating, the remaining liquid loses internal energy and cools. Thus (1) and (2) are correct. Answer: A.

Question 4

The pressure of a fixed mass of an ideal gas at $10\text{ }^{\circ}\text{C}$ is $2 \times 10^5\text{ N m}^{-2}$. If the volume of the gas is reduced to half of its original volume and its temperature is increased to $100\text{ }^{\circ}\text{C}$, what would the pressure be?

- A. $1.00 \times 10^5\text{ N m}^{-2}$
- B. $1.32 \times 10^5\text{ N m}^{-2}$
- C. $4.00 \times 10^5\text{ N m}^{-2}$
- D. $5.27 \times 10^5\text{ N m}^{-2}$

Official answer: D

For a fixed mass of ideal gas, pV/T is constant with T in kelvin. Thus $p_2 = p_1(V_1/V_2)(T_2/T_1) = 2 \times 10^5 \times 2 \times 373/283 = 5.27 \times 10^5\text{ Pa}$. Using 100/10 would incorrectly use Celsius temperatures in the gas law. Answer: D.

Question 5

Which of the following statements about the motion of any two objects is correct?

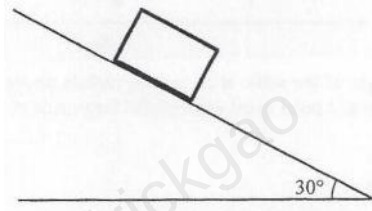
- A. The object that takes a shorter time to complete the same path must have greater average speed.
- B. The object that travels a greater distance in 1 s must have greater average velocity.
- C. The object with greater velocity must have greater acceleration.
- D. If the two objects have the same acceleration, they must be moving in the same direction.

Official answer: A

Average speed is total distance divided by elapsed time. Completing the same path in less time therefore means greater average speed. Average velocity depends on displacement, not distance; velocity does not determine acceleration; and acceleration direction need not match velocity direction. Answer: A.

Question 6

A block is released from rest on an inclined plane as shown. The inclined plane makes an angle of 30° to the horizontal. The block moves with uniform acceleration, and travels a distance of 1 m in the first 3 s. Determine the acceleration of the block.



- A. 0.22 m s^{-2}
- B. 0.33 m s^{-2}
- C. 4.91 m s^{-2}
- D. Cannot be determined as the frictional force acting on the block is unknown.

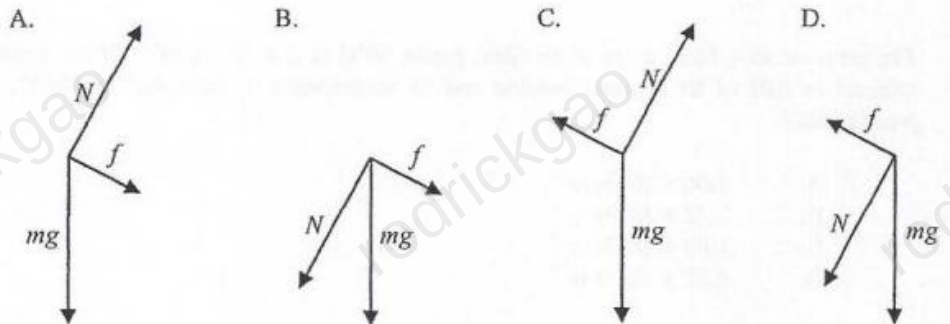
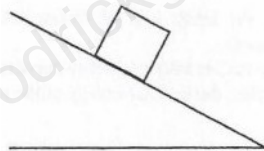
Official answer: A

Since the block starts from rest with uniform acceleration, $s = ut + \frac{1}{2}at^2$ gives $1 = \frac{1}{2}a(3)^2$. Thus $a = 2/9 = 0.22 \text{ m s}^{-2}$ down the slope. Friction need not be known because the displacement and time already determine the acceleration. Answer: A.

Question 7

A block of mass m stays at rest on a rough inclined plane as shown.

Which of the following diagrams correctly shows the forces acting on the block? (N is the normal reaction from the inclined plane, and f is the frictional force between the block and the plane.)



Official answer: C

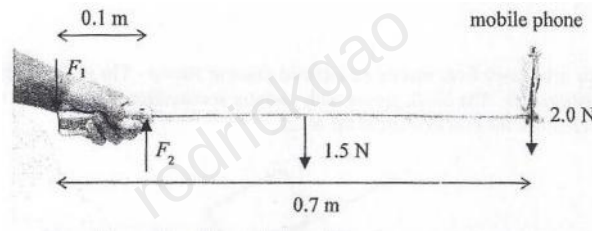
Weight mg acts vertically downward. The normal reaction points perpendicular to and away from the plane. Static friction acts uphill, opposing the tendency to slide downhill. The three forces balance because the block is at rest. Only diagram C has all three directions correct. Answer: C.

Question 8

Selfie sticks are popular nowadays. A uniform selfie stick of length 0.7 m is held horizontally as shown.

Assume that the forces required to hold the selfie stick by the hand are represented by F_1 and F_2 , and F_1 and F_2 are perpendicular to the stick.

It is given that the weight of the selfie stick and the mobile phone are 1.5 N and 2.0 N respectively. Taking the mobile phone as a point mass, estimate the magnitude of F_2 .



- A. 3.5 N
- B. 19.3 N
- C. 35 N
- D. Cannot be determined as F_1 is unknown.

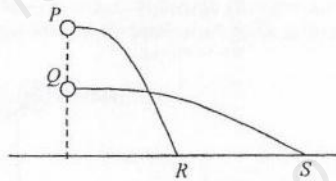
Official answer: B

Take moments about the point where F_1 acts, so F_1 contributes no moment. The uniform stick's weight acts at its midpoint, 0.35 m from this point. Thus $F_2(0.10) = 1.5(0.35) + 2.0(0.70)$, giving $F_2 = 19.25 \text{ N} \approx 19.3 \text{ N}$. Answer: B.

Question 9

Marbles P and Q of the same mass are shot horizontally. They hit the horizontal ground at points R and S respectively as shown. Neglect air resistance.

Which of the following statements is INCORRECT?



- A. The initial speed of marble P is smaller than that of marble Q.
- B. The time of flight of marble P is shorter than that of marble Q.
- C. The potential energy loss of marble P is greater than that of marble Q.
- D. The acceleration of marbles P and Q is the same during the flight.

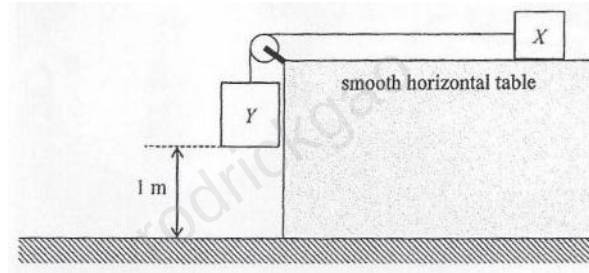
Official answer: B

For horizontal projection, flight time is $t = \sqrt{2h/g}$. P starts higher, so $t_P > t_Q$; statement B is incorrect. P also travels a shorter horizontal range despite its longer flight, so its initial horizontal speed is smaller. Its potential-energy loss is larger, while both accelerations are g downward. Answer: B.

Question 10

Blocks X and Y are connected by a light inextensible string passing over a fixed frictionless light pulley as shown. The masses of X and Y are 0.5 kg and 1 kg respectively. Initially, Y is 1 m above the ground and the string is taut. The system is then released from rest.

What is the speed of Y just before it reaches the ground? (Take $g = 9.81 \text{ m s}^{-2}$)



- A. 3.62 m s^{-1}
- B. 4.43 m s^{-1}
- C. 6.26 m s^{-1}
- D. 9.81 m s^{-1}

Official answer: A

The smooth table and ideal pulley allow mechanical-energy conservation. Both blocks have the same speed v because the string is taut and inextensible. Thus $(1)(9.81)(1) = \frac{1}{2}(0.5+1)v^2$, giving $v = 3.62 \text{ m s}^{-1}$. X also gains kinetic energy, so Y is not in free fall. Answer: A.

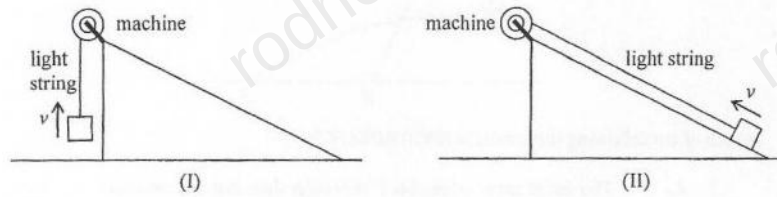
Question 11

A machine is fixed at the top of a smooth inclined plane. Two methods, (I) and (II), are used to lift a block from the ground to the top of the inclined plane by the machine.

- (I) Pull the block vertically upward at a uniform speed v .
 (II) Pull the block up along the inclined plane at the same uniform speed v .

Which of the following statements correctly compare(s) the two methods?

- (1) The tension in the string is the same.
 (2) The average output power of the machine is the same.
 (3) The work done by the machine on the block is the same.



- A. (1) only
 B. (3) only
 C. (1) and (2) only
 D. (2) and (3) only

Official answer: B

At constant speed, the vertical lift requires tension mg , while the incline requires $mg \sin \theta$. At equal speed, output power Fv is therefore smaller on the incline. Both methods raise the same block through the same height without friction, so the work is mgh in either case. Only (3) is correct. Answer: B.

Question 12

Players of “bubble soccer” wear air-filled plastic “bubbles” as shown.

Which of the following statements best explains why the bubble can reduce the chance of injury during a collision?



- A. The bubble increases the mass of the player, thus the momentum of the player increases.
- B. The bubble increases the air resistance acting on the player.
- C. The bubble lengthens the impact time during a collision.
- D. Like a balloon, the bubble provides a lifting force to the player.

Official answer: C

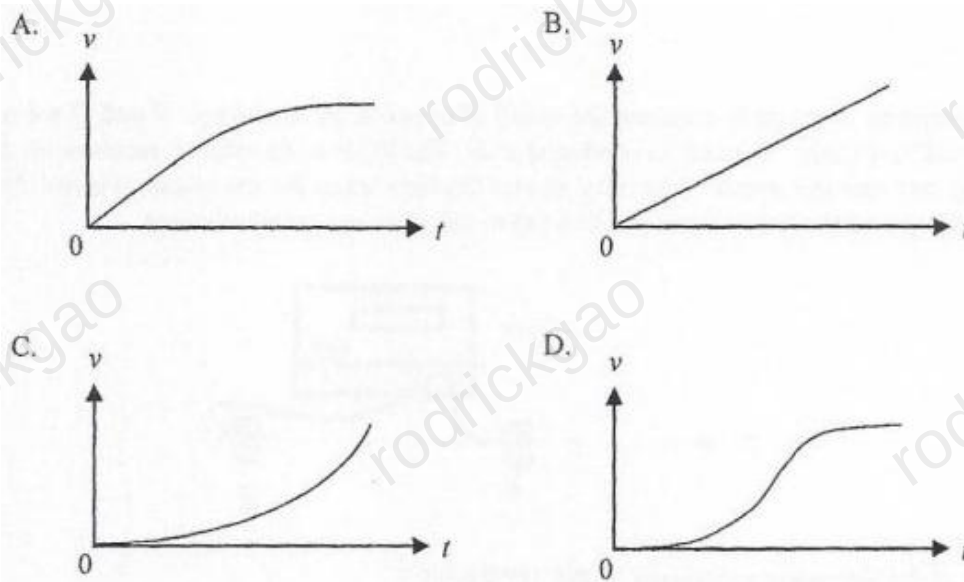
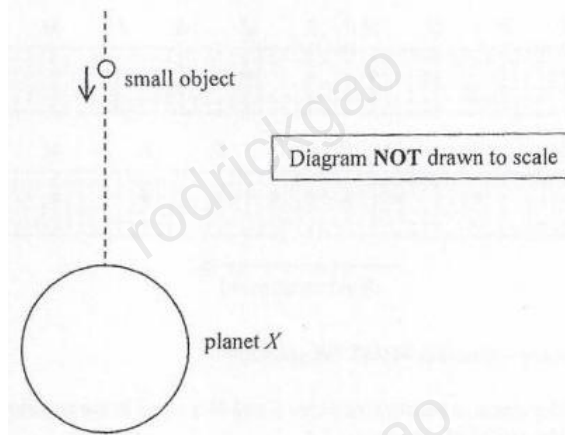
Impulse equals change of momentum: $F_{\text{avg}} \Delta t = \Delta p$. For a comparable momentum change in a collision, the deformable air-filled bubble increases the impact time and reduces the average impact force on the player.

Answer: C.

Question 13

A small object is released from rest at a point very far away from a planet X. The object then starts moving towards X. X does not have an atmosphere. Neglect the effect of other celestial bodies.

Which of the following graphs best shows the variation of the velocity v of the object with time t before it hits X?



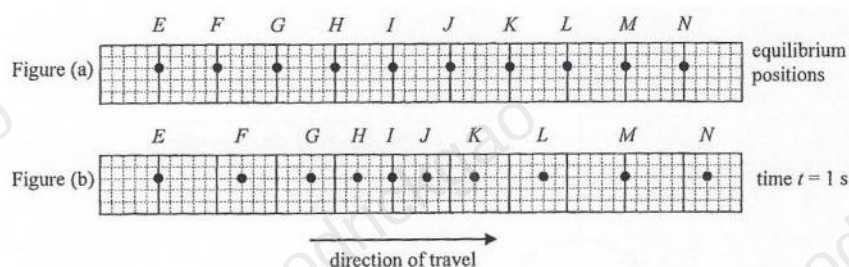
Official answer: C

As the object approaches the planet, r decreases and gravitational acceleration GM/r^2 increases. The slope of the velocity–time graph therefore increases, starting very small far from the planet. With no atmosphere, there is no drag-induced terminal speed. This is graph C. Answer: C.

Question 14

Figure (a) shows the equilibrium positions of particles E to N in a medium. At time $t = 0$, a longitudinal wave starts travelling from left to right. At time $t = 1$ s, the positions of the particles are shown in Figure (b).

Which of the following statements **MUST BE** correct?



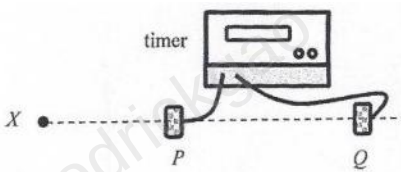
- A. The distance between particles F and N is equal to the wavelength of the wave.
- B. The period of the wave is 1 s.
- C. Particle E is always at rest.
- D. Particle I is momentarily at rest at $t = 1$ s.

Official answer: A

F and N occupy equivalent positions in successive cycles of the displacement pattern, separated by one wavelength. A single snapshot does not establish a period of 1 s. E being at equilibrium at one instant does not mean it never moves, and I is crossing equilibrium rather than at maximum displacement. Answer: A.

Question 15

An experiment is set up to measure the speed of sound in air as shown. P and Q are two microphones connected to a timer. A sound is produced at X. The timer starts when P receives the sound, and stops when Q receives the sound. The timer shows the time taken for the sound to travel from P to Q. The distance PQ and the time shown can be used to calculate the speed of sound. Which of the following statements is INCORRECT?



- A. X, P and Q must be along the same straight line.
- B. The percentage error in the time measured will increase if the distance PQ is reduced.
- C. The speed of sound determined should be independent of the distance between X and P.
- D. The distance PQ must be equal to an integral multiple of wavelengths of the sound produced at X.

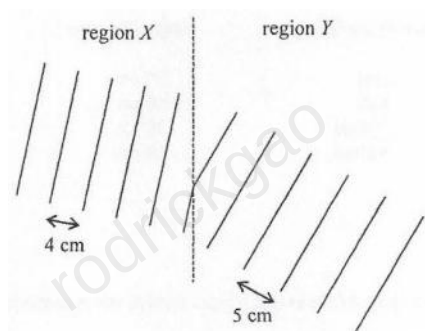
Official answer: D

This is a time-of-flight measurement, $v = PQ/\Delta t$, not a phase comparison or standing-wave experiment. PQ need not be an integral number of wavelengths. Reducing PQ shortens Δt and increases percentage timing uncertainty for a fixed absolute uncertainty. Collinear X, P and Q make the path difference equal to PQ. Answer: D.

Question 16

The figure shows plane water waves travelling from region X to region Y. The wavelengths of the water waves in regions X and Y are 4 cm and 5 cm respectively.

Which of the following statements is correct?



- A. The speed of the water waves in region X is higher than that in region Y.
- B. The direction of travel of the water waves bends towards the normal as they enter region Y.
- C. The frequency of the water waves is the same in both regions.
- D. If plane water waves of wavelength 5 cm travel from region Y to region X, the wavelength becomes 6 cm after the waves enter region X.

Official answer: C

The source fixes the frequency, which remains unchanged across the boundary. From $v = f\lambda$, $v_Y/v_X = 5/4$, so the wave is faster in Y and bends away from the normal. On returning to X at the same frequency, its wavelength would be 4 cm, not 6 cm. Answer: C.

Question 17

In which of the following situations MUST the direction of travel of a wave change?

- (1) when a wave is reflected by a barrier
- (2) when a wave enters from one medium to another medium
- (3) when a wave travels through a gap smaller than its wavelength

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: B

Reflection changes the direction of propagation. A gap smaller than the wavelength produces strong diffraction and spreading into new directions. Entering another medium need not change direction: normal incidence is a counterexample. Hence (1) and (3) are correct. Answer: B.

Question 18

Two musical notes of the same pitch and loudness are produced by two different musical instruments. They sound different to the human ears because they have different

- A. amplitudes.
- B. phases.
- C. wave speeds.
- D. waveforms.

Official answer: D

Different instruments can produce the same fundamental frequency and comparable amplitude but different harmonic content. Their waveforms, and hence timbre, differ. The common medium does not give them different sound speeds. Answer: D.

Question 19

When an object is placed 30 cm in front of a concave lens, an image is formed 20 cm away from the lens. If the concave lens is replaced by a convex lens of the same focal length and the object distance remains unchanged, which of the following descriptions about the image formed is correct?

- A. Nature: real; image distance: 20 cm
- B. Nature: real; image distance: 60 cm
- C. Nature: virtual; image distance: 20 cm
- D. Nature: virtual; image distance: 60 cm

Official answer: D

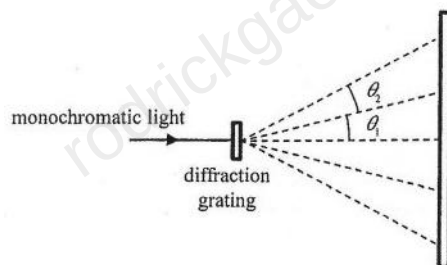
Use $1/f = 1/u + 1/v$, with a virtual image assigned negative v . For the concave lens, $1/f = 1/30 - 1/20 = -1/60$, so $f = -60$ cm. Replacing it by a convex lens gives $f = +60$ cm and $1/v = 1/60 - 1/30 = -1/60$. The new image is virtual, 60 cm from the lens. Answer: D.

Question 20

The figure below shows some of the bright fringes formed when monochromatic light passes through a diffraction grating.

Which of the following is/are correct?

- (1) $\theta_1 = \theta_2$.
- (2) The maximum order of bright fringe is 4 if $\theta_1 = 20^\circ$.
- (3) θ_1 will decrease if the experiment is performed in water but not in air.



- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: B

Grating maxima obey $d \sin \theta = n\lambda$. Equal increments of order do not give equal angular gaps, so θ_1 and θ_2 need not be equal. If the first-order angle is 20° , $d/\lambda = 1/\sin 20^\circ = 2.92$, so the maximum order is 2, not 4. In water the wavelength is shorter at unchanged frequency, reducing θ_1 . Only (3) is correct. Answer: B.

Question 21

If the speed of sound in water is x and the speed of light in water is y , which of the following is correct?

- A. Speed of sound in air $> x$; speed of light in air $> y$
- B. Speed of sound in air $> x$; speed of light in air $< y$
- C. Speed of sound in air $< x$; speed of light in air $> y$
- D. Speed of sound in air $< x$; speed of light in air $< y$

Official answer: C

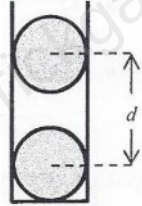
Sound travels faster in water than in air, so its speed in air is less than x . Light travels slower in water than in air, so its speed in air is greater than y . Answer: C.

Question 22

In the figure, two charged conducting spheres of the same mass m are put in a vertical plastic cylinder. The inner wall of the cylinder is smooth. The spheres are separated by a distance d and remain in equilibrium.

Which of the following statements **MUST BE** correct?

- (1) Both spheres carry positive charges.
- (2) The amount of charges on the two spheres are the same.
- (3) The separation d depends on m .



- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: B

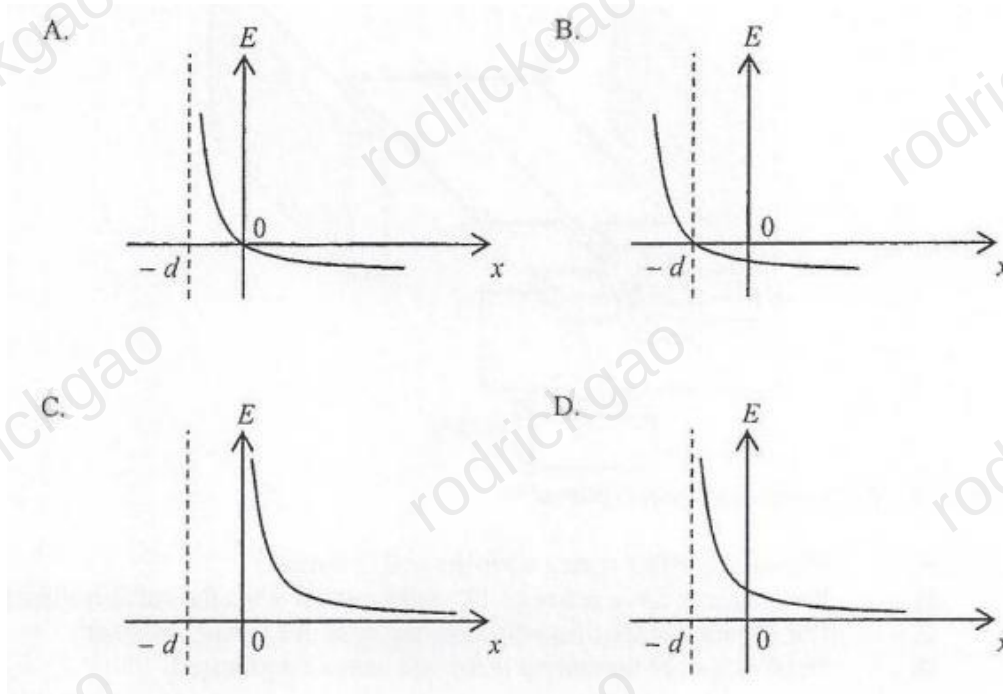
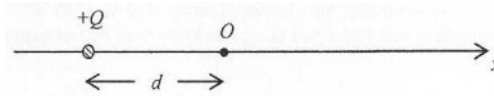
For the upper sphere, upward electric repulsion balances mg . The spheres need like charges, but both could be negative and their charge magnitudes need not be equal. For fixed charges the equilibrium separation depends on weight; in the point-charge approximation, $k|q_1q_2|/d^2 = mg$. Only (3) is required. Answer: B.

Question 23

A point charge $+Q$ is fixed at a distance d away from the origin O as shown.

Which of the following graphs best represents the variation of the electric field strength E along the x -axis?

(Take the electric field pointing to the right as positive.)

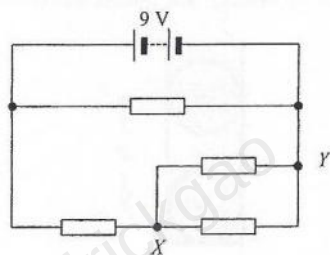


Official answer: D

The charge lies at $x = -d$. On its right, E points right and is positive, with $E = kQ/(x+d)^2$. It diverges as x approaches $-d$ from the right and tends to zero as x increases; it is finite and nonzero at $x = 0$. The shown branch is graph D. Answer: D.

Question 24

In the circuit, all resistors are identical. The internal resistance of the battery can be neglected. What is the potential difference between X and Y?



- A. 1.5 V
- B. 3.0 V
- C. 4.5 V
- D. 6.0 V

Official answer: B

The two resistors between X and Y are in parallel, giving $R/2$. This combination is in series with the remaining lower-branch resistor R across 9 V. Hence $V_{XY} = 9(R/2)/(R+R/2) = 3.0$ V. The separate upper resistor does not change the ideal battery voltage. Answer: B.

Question 25

Which of the following statements about the use of a fuse is correct?

- A. A fuse should be installed in the neutral wire.
- B. A fuse is not required in an electrical appliance with double insulation.
- C. A 5 A fuse is suitable for a heater of rating “220 V, 1500 W”.
- D. The melting point of a fuse should be lower than that of copper.

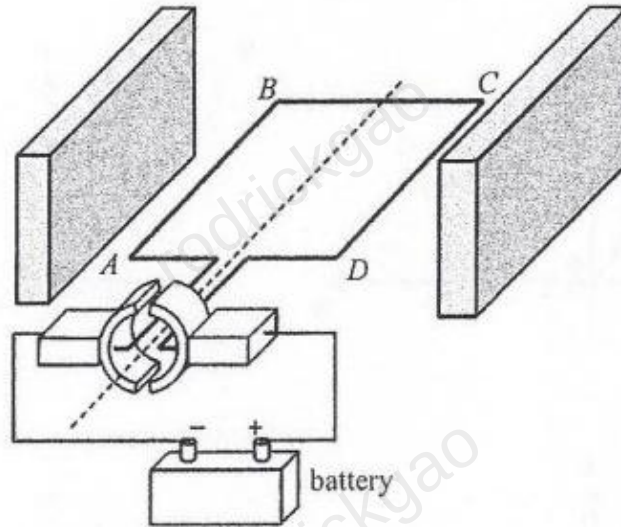
Official answer: D

A fuse must melt before the copper wiring overheats dangerously, so it uses a lower-melting-point material. It belongs in the live wire, and double insulation does not replace overcurrent protection. The heater's normal current is $1500/220 = 6.82$ A, above a 5 A fuse rating. Answer: D.

Question 26

The figure shows a simple d.c. motor; the coil ABCD is mounted between the poles of two slab-shaped magnets.

Which of the following statements is correct?



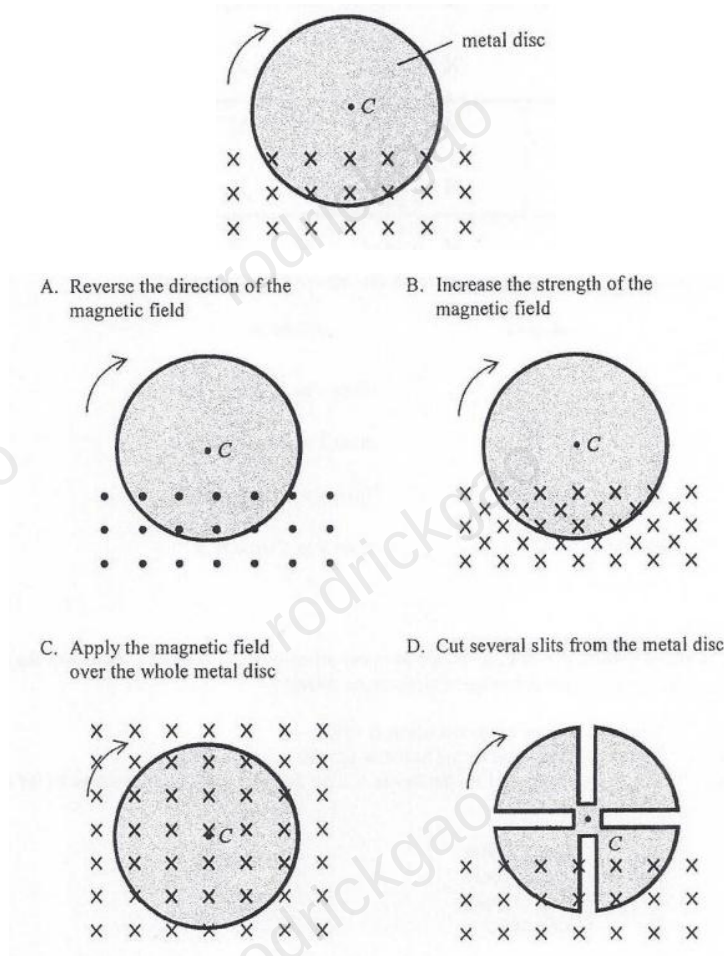
- A. The turning effect is zero when the coil is vertical.
- B. The magnetic force acting on BC is the greatest when the coil is horizontal.
- C. The direction of the magnetic force acting on AB remains constant.
- D. The direction of the current in the coil remains unchanged.

Official answer: A

When the coil is vertical, its magnetic moment is parallel to the field and the torque is zero. In a simple d.c. motor the split-ring commutator reverses the coil current every half-turn. Thus the current and force on a given active side do not remain unchanged throughout a revolution. Answer: A.

Question 27

A metal disc is rotating about its centre C with constant speed. Part of the metal disc is inside a uniform magnetic field pointing into the paper as shown. An eddy current flows in the metal disc. After which of the following changes will the eddy current increase?



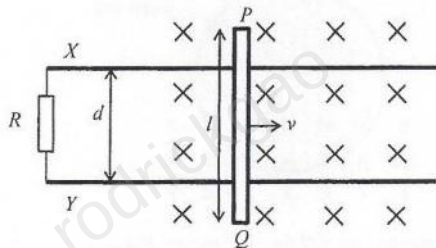
Official answer: B

Increasing magnetic-field strength increases the rate of flux change as parts of the rotating disc enter and leave the field, producing a larger induced emf and eddy current. Reversing the field reverses current direction, not its magnitude. Slits obstruct current loops; a uniform field over the entire disc removes the entering/leaving-field effect. Answer: B.

Question 28

A metal rod PQ of length l is moving along smooth horizontal metal rails X and Y with constant speed v in a uniform magnetic field of magnetic field strength B pointing into the paper. The metal rails X and Y are separated by a distance d and are connected to a resistor of resistance R as shown.

Which of the following descriptions about the induced current is correct?



- A. Magnitude Blv/R ; direction from X to Y through R
- B. Magnitude Blv/R ; direction from Y to X through R
- C. Magnitude Bdv/R ; direction from X to Y through R
- D. Magnitude Bdv/R ; direction from Y to X through R

Official answer: C

The effective moving conductor length between the rails is d , not the full rod length l . Thus $\mathcal{E} = Bdv$ and $I = Bdv/R$, neglecting rail and rod resistance. Moving right increases inward flux, so the induced current is anticlockwise and flows through R from X to Y. Answer: C.

Question 29

A heater of resistance $100\ \Omega$ is connected to the mains supply. The r.m.s. voltage of the mains supply is 110 V . Which of the following statements are correct?

- (1) The peak voltage across the heater is 156 V .
- (2) The power dissipated by the heater is 121 W .
- (3) The power dissipated by the heater will be doubled if the r.m.s. voltage of the mains supply doubles.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: A

For a sinusoidal supply, $V_{\text{peak}} = \sqrt{2} V_{\text{rms}} = \sqrt{2}(110) \approx 156\text{ V}$. Average power is $V_{\text{rms}}^2/R = 110^2/100 = 121\text{ W}$. Doubling the r.m.s. voltage quadruples this power for unchanged resistance, not doubles it. Hence (1) and (2) are correct. Answer: A.

Question 30

The input terminal of a transformer is connected to the 220 V mains supply. Ten identical light bulbs are connected in parallel to the output terminal of the transformer. All the light bulbs are working at their rated values of “3 V, 1.5 W”. If the efficiency of the transformer is 70%, what is the current drawn from the mains supply?

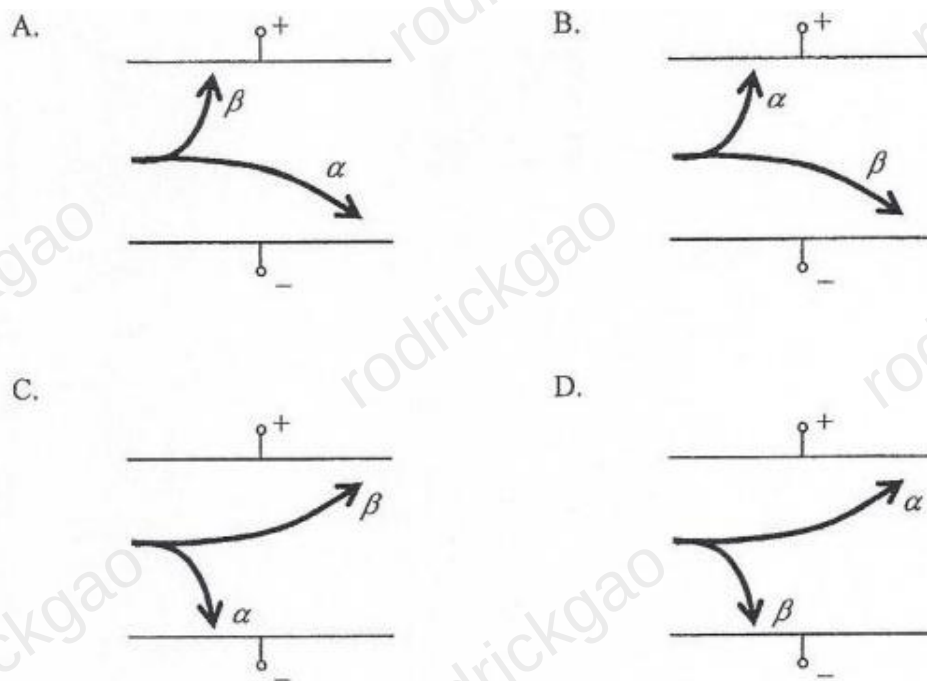
- A. 0.007 A
- B. 0.048 A
- C. 0.068 A
- D. 0.097 A

Official answer: D

Total output power is $10 \times 1.5 = 15$ W. Efficiency $\eta = P_{\text{out}}/P_{\text{in}}$ gives $P_{\text{in}} = 15/0.70 = 21.43$ W. The mains current is $I_{\text{in}} = P_{\text{in}}/220 = 0.0974$ A ≈ 0.097 A. Use input voltage with input power, not with output power. Answer: D.

Question 31

Which of the following diagrams best shows the deflection of α and β particles in a uniform electric field in vacuum?



Official answer: A

The field points from the positive upper plate to the negative lower plate. Positive α particles deflect downward, while negative β particles deflect upward. In the standard radiation comparison shown, β particles are deflected more strongly; diagram A has the correct directions and relative bending. Answer: A.

Question 32

Which of the following statements about β particles and γ rays is correct?

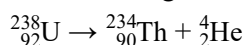
- A. Only β particles can ionize air particles.
- B. Only γ rays can travel through vacuum.
- C. Both of them can be detected by a photographic film.
- D. Both of them carry charge.

Official answer: C

Both β radiation and γ radiation can expose photographic film and can ionize matter. Both can travel through vacuum. β particles are charged, whereas γ photons are uncharged. Only statement C is correct. Answer: C.

Question 33

The following shows the decay of uranium-238 ($^{238}_{92}\text{U}$):



Given that:

mass of $^{238}_{92}\text{U} = 238.05079 \text{ u}$

mass of $^{234}_{90}\text{Th} = 234.04363 \text{ u}$

mass of $^4_2\text{He} = 4.00260 \text{ u}$

Which of the following statements is/are correct?

- (1) The temperature required to start the decay is about 10^7 K .
- (2) The energy released in the decay of one uranium-238 nucleus is 4.25 MeV.
- (3) All the energy released in the decay becomes the kinetic energy of ^4_2He .

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

Official answer: B

The mass decrease is $238.05079 - 234.04363 - 4.00260 = 0.00456 \text{ u}$. Released energy is $\Delta mc^2 \approx 0.00456 \times 931.5 = 4.25 \text{ MeV}$. Radioactive α decay is spontaneous and does not require a temperature of 10^7 K . Momentum conservation gives recoil kinetic energy to the daughter nucleus, so not all released energy goes to the α particle. Answer: B.